

9. Compare Linear and Polynomial Regression using Python

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression

from sklearn.preprocessing import PolynomialFeatures

from sklearn.metrics import mean_squared_error, r2_score

X = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]).reshape(-1, 1)

y = np.array([2, 4, 7, 11, 16, 22, 29, 37, 46, 56])

linear_model = LinearRegression()

linear_model.fit(X, y)

y_linear = linear_model.predict(X)

poly = PolynomialFeatures(degree=2)

X_poly = poly.fit_transform(X)

poly_model = LinearRegression()

poly_model.fit(X_poly, y)

y_poly = poly_model.predict(X_poly)

print("Linear Regression")

print("MSE:", mean_squared_error(y, y_linear))

print("R2 Score:", r2_score(y, y_linear))

print("\nPolynomial Regression")

print("MSE:", mean_squared_error(y, y_poly))

print("R2 Score:", r2_score(y, y_poly))

plt.scatter(X, y, label="Actual Data")

plt.plot(X, y_linear, label="Linear Regression")

plt.plot(X, y_poly, label="Polynomial Regression")

plt.xlabel("X")
```

```
plt.ylabel("Y")  
plt.title("Linear vs Polynomial Regression")  
plt.legend()  
plt.show()
```

```
Linear Regression  
MSE: 13.200000000000008  
R2 Score: 0.9574468085106382
```

```
Polynomial Regression  
MSE: 1.4777336907052603e-28  
R2 Score: 1.0
```

